Diffeomorphic Temporal Alignment Nets

Supplemental Material

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33rd Conference on Neural Information Processing Systems (NeurIPS 2019), Vancouver, Canada.

1 Temproal Transformer Nets

1.1 Model Architecture

As mentioned in our paper, a Temporal Transformer (TT) layer (Figure 1) is a variant of the Spatial Transformer layer [2], and is consisted of 3 modules:

1. A localization network. For an input signal, $\boldsymbol{U}$, the localization network, f_{loc}, regresses the warp's parameters such that $f_{\text{loc}}(\boldsymbol{U}) = \boldsymbol{\theta}$.

2. A parameterized grid generator. This generator creates a discrete 1D grid of length M (where M is the length of $\boldsymbol{U}$), $G = (p_m)_{m=1}^{M} \subset [-1, 1]$, of evenly-spaced points.

3. A differentiable time-series resampler. The output signal, $\boldsymbol{V}$, is computed by interpolating the values of $\boldsymbol{V}$ at $T^{\boldsymbol{\theta}}(G)$ from $\boldsymbol{U}$, as explained below.

Let $p_{i,m}^{\text{warped}} = T^{\boldsymbol{\theta}_i}(p_m)$. The discrete-time i-th aligned signal is:

$$\boldsymbol{V}_i = (\boldsymbol{V}_{i,m})_{m=1}^{M} = (\boldsymbol{V}_{i,1}, \ldots, \boldsymbol{V}_{i,M})\,. \tag{1}$$

Note that due to the need to resample the signal, rather than having $\boldsymbol{V}_i = \boldsymbol{U}_i \circ T^{\boldsymbol{\theta}_i}$, we must also account for the resampling kernel. For the popular linear kernel, we obtain (based on [2]),

$$\boldsymbol{V}_{i,m} = \sum_{m'=1}^{M} \boldsymbol{U}_{i,m'} \max(0, 1 - |p_{i,m}^{\text{warped}} - m'|)\,. \tag{2}$$

To propagate the loss to the localization network, the resampling kernel must be differentiable, which is the case for the linear kernel used in this paper.

1.2 Derivatives

We provide the derivatives for the 1D Temporal Alignment Network (based on [2]). The derivative w.r.t. the parameterization of the warp family (*i.e.* the CPAB gradient) is discussed in the main paper.

$$\frac{\partial \boldsymbol{V}_{i,m}}{\partial \boldsymbol{U}_{i,m'}} = \max(0, 1 - |p_{i,m}^{\text{warped}} - m'|) \tag{3}$$

$$\frac{\partial \boldsymbol{V}_{i,m}}{\partial (p_{i,m}^{\text{warped}})} = \sum_{m'=1}^{M} \boldsymbol{U}_{i,m'} \begin{cases} 0 & \text{if} \quad |m' - p_{i,m}^{\text{warped}}| \geq 1 \\ 1 & \text{if} \quad m' \geq p_{i,m}^{\text{warped}} \\ -1 & \text{if} \quad m' < p_{i,m}^{\text{warped}} \end{cases}. \tag{4}$$

Where $\boldsymbol{V}_{i,m}$ is the ith warped signal at time point m, $\boldsymbol{U}_{i,m'}$ is the input signal at time point m' and $p_{i,m}^{\text{warped}}$ is the mth point of the sampling grid. The generalization of these results to multichannel time series is straightforward and thus omitted.

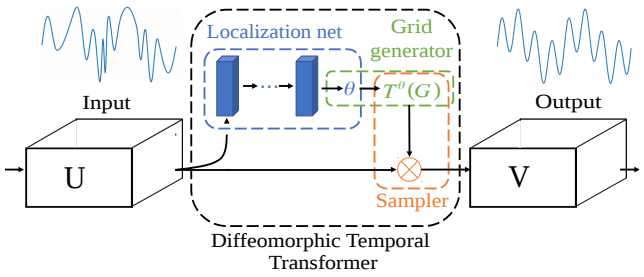


Figure 1: The Diffeomorphic Temporal Transformer module. Figure adapted with permission from [4].

Table 1: Train and Test syntactic data experiments with an added Gaussian noise. The left tables corresponds to $\sigma = 0.01$ while the right one corresponds to $\sigma = 0.1$.

$\sigma = 0.01$		
Train set variance		
Dataset	Baseline	DTAN4
$K = 32$	0.485	**0.063**
$K = 16$	0.521	**0.110**
$K = 8$	0.536	**0.125**
Test set variance		
$K = 32$	0.468	**0.097**
$K = 16$	0.512	**0.173**
$K = 8$	0.532	**0.185**

$\sigma = 0.1$		
Train set variance		
Dataset	Baseline	DTAN4
$K = 32$	0.502	**0.091**
$K = 16$	0.545	**0.147**
$K = 8$	0.563	**0.177**
Test set variance		
$K = 32$	0.501	**0.141**
$K = 16$	0.533	**0.161**
$K = 8$	0.544	**0.193**

2 Synthetic Data Experimental Settings

The synthetic data generation procedure is as follows. First, we generated 4 input signals, each is a linear combination of 2 sinus signals of different frequencies. Next, we generated n warped signals for each input signal in the following manner: we sampled a K-bin histogram from a symmetric K-dimensional Dirichlet distribution. For each such histogram, we calculated its Cumulative Distribution Function (CDF). Since a CDF is a monotonically non-decreasing function, it can be used as a (non-differentiable) warp from the unit interval into itself, while keeping the endpoints fixed. Thus, each CDF corresponds to random warp. We use these warps to generate the synthetic warp data. Note that as K increases, each sample is more likely to be close to the uniformly distribution, which in turn means a close-to-linear CDF, *i.e.*, almost the identity warp. The value of n was set to 250 unless stated otherwise. The network was trained for 2500 epochs with an Adam optimizer [3], learning rate $\eta = 10^{-4}$. The parameters of the smoothness prior were $\lambda_{val} = 0.01$ and $\lambda_{smooth} = 1$. The Partition, Ω, was set such that $\dim(\boldsymbol{\theta}) = 32$. The Dirichlet distribution's K values are either 8, 16, or 32, while the value of α (the parameter of the symmetric Dirichlet distribution) was fixed to $\alpha = 3$.

The train set contained 800 signals (200 per class) where the test set contained 200 signals. We used a 60-20-20% train, validation, and test split. After training, we computed the joint alignment of these sets by simply passing each through the network. As such, computing the average signal of each class is given by the arithmetic mean of each class of the aligned data.

Synthetic Data with an Additive Gaussian Noise. We expended our synthetic experiment (reported in the paper) by adding noise to increase the difficulty of the input data. We generated the warped signal similarly to the previous experiment, and then added an additional noise from a Gaussian distribution of mean $\mu = 0$ and a standard deviation of either $\sigma = 0.1$ or $\sigma = 0.01$ to the signal's values. Results are reported in Table 1.

3 Computing Infrastructures

We have used the following infrastructures in our experiments: Intel(R) Core(TM) i7-7740X CPU @ 4.30GHz, 8 cores, 16gb RAM with an Nvidia GeForce GTX 1080 graphic card.

4 Regularization effect on DTAN Joint Alignment

4.1 Regularization Effect

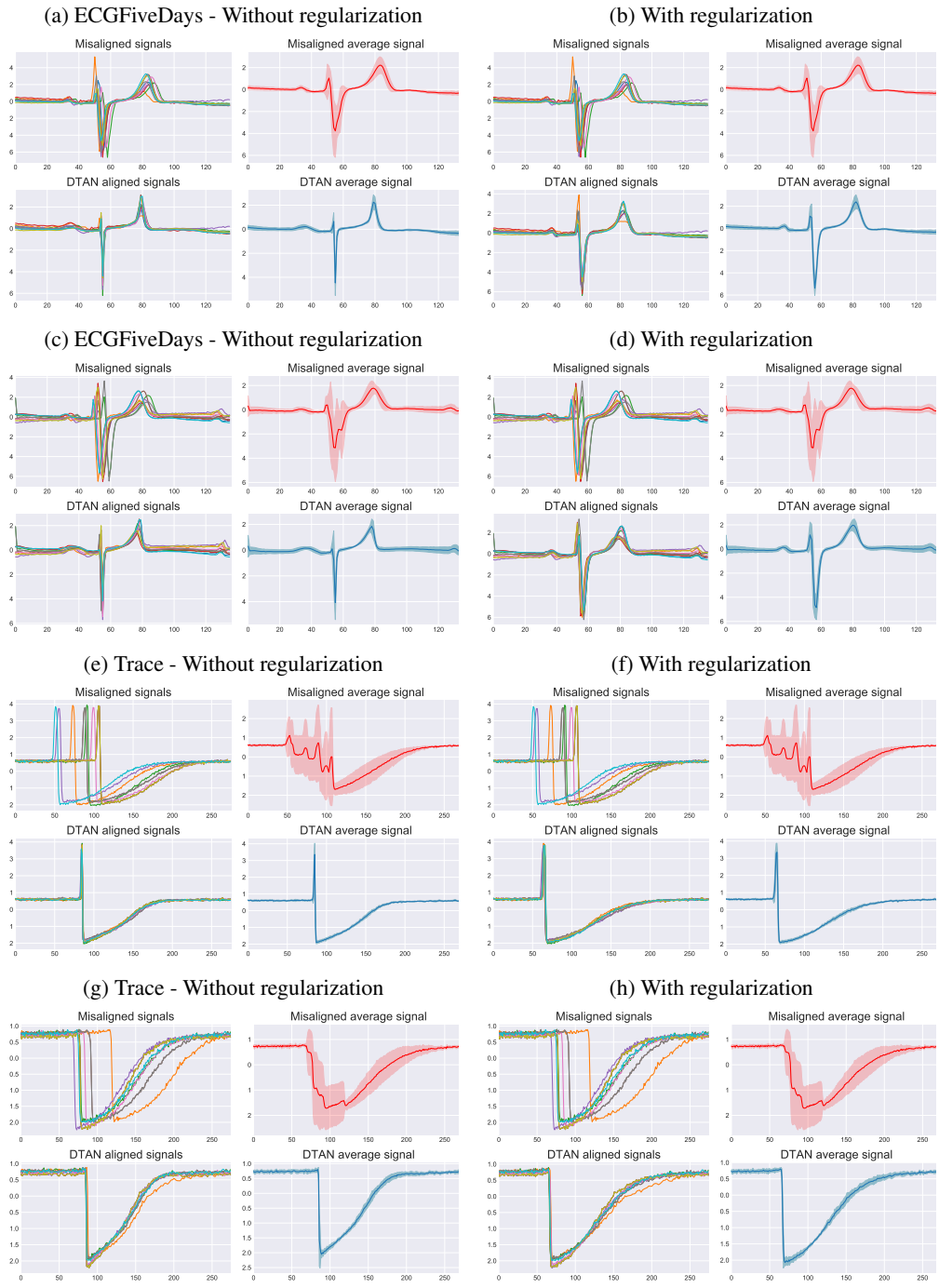


Figure 2: Regularization Effect on the ECGFiveDays(a)-(d) and Trace (e)-(h) datasets. Left: without the regularization term (F_{reg}). Right: with F_{reg} ($\lambda_{var} = 0.1, \lambda_{smooth} = 1$). Since DTAN framework is unsupervised, F_{data} could be minimized by unrealistically-large deformations; F_{reg} prevents that.

4.2 With a Zero Boundary Condition on the CPA Velocity Field

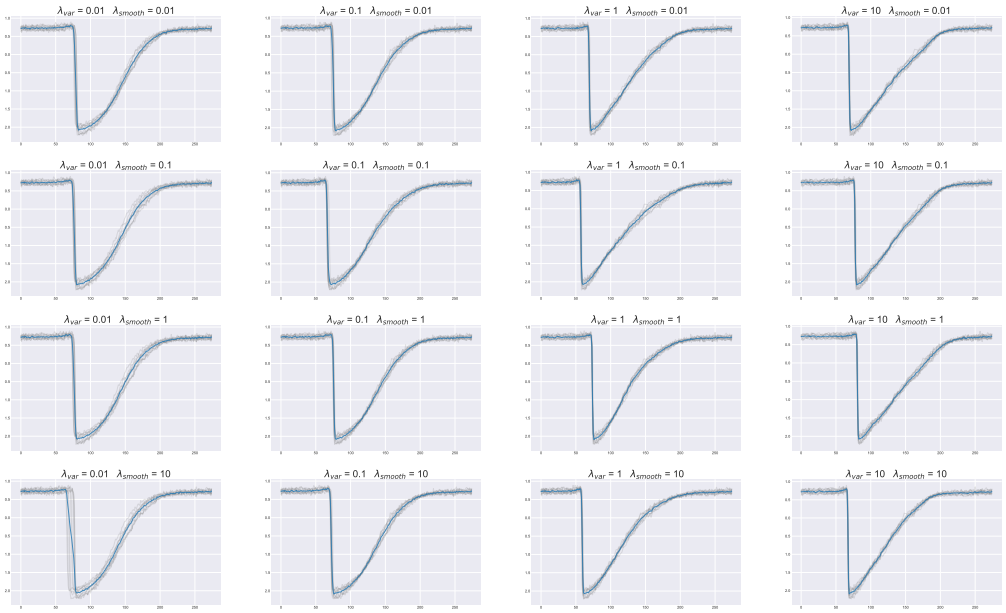


Figure 3: Joint alignment of 10 randomly-chosen samples of a given class from the Trace dataset. $\lambda_{\text{var}}, \lambda_{\text{smooth}} \in [0.01, 0.1, 1, 10]$. $\dim(\boldsymbol{\theta}) = 8$.

4.3 Without a Zero Boundary Condition on the CPA Velocity Field

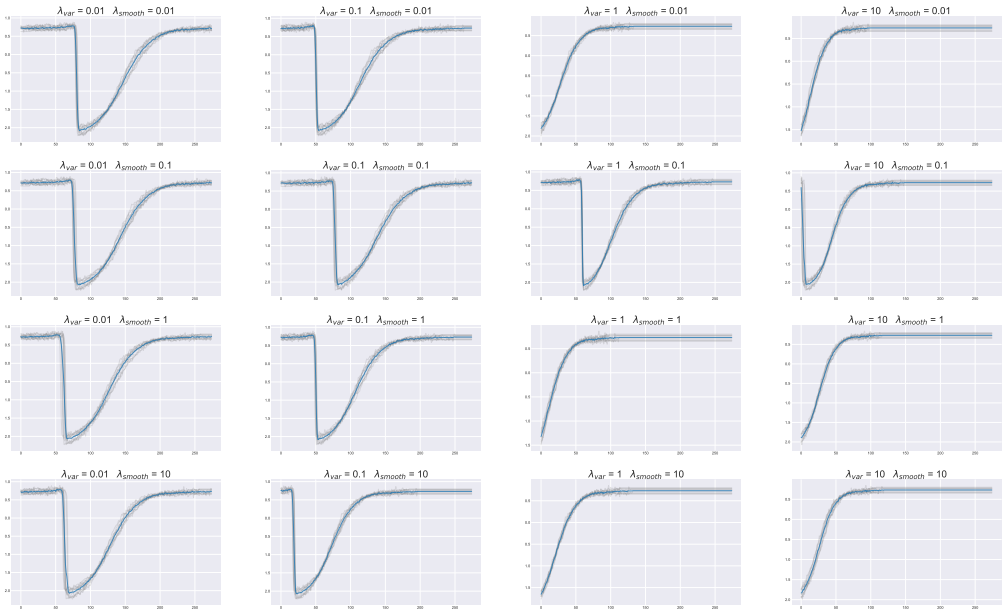


Figure 4: Removing the zero-boundary condition. Joint alignment of 10 randomly-chosen test samples of a given class from the Trace dataset. $\lambda_{\text{var}}, \lambda_{\text{smooth}} \in [0.01, 0.1, 1, 10]$. $\dim(\boldsymbol{\theta}) = 8$.

5 Additional Alignment Results of Test Data

5.1 Within-class Variance Reduction

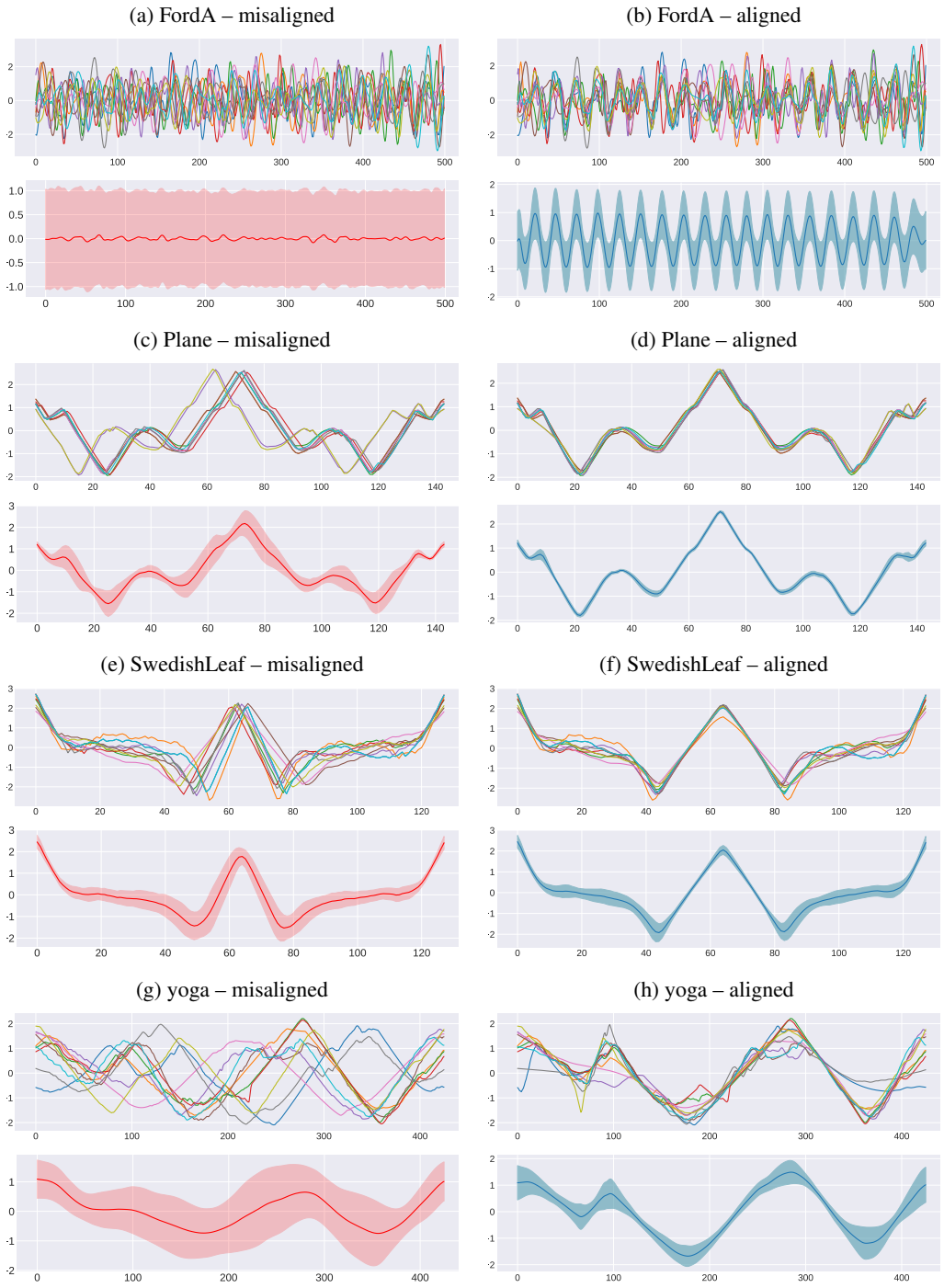
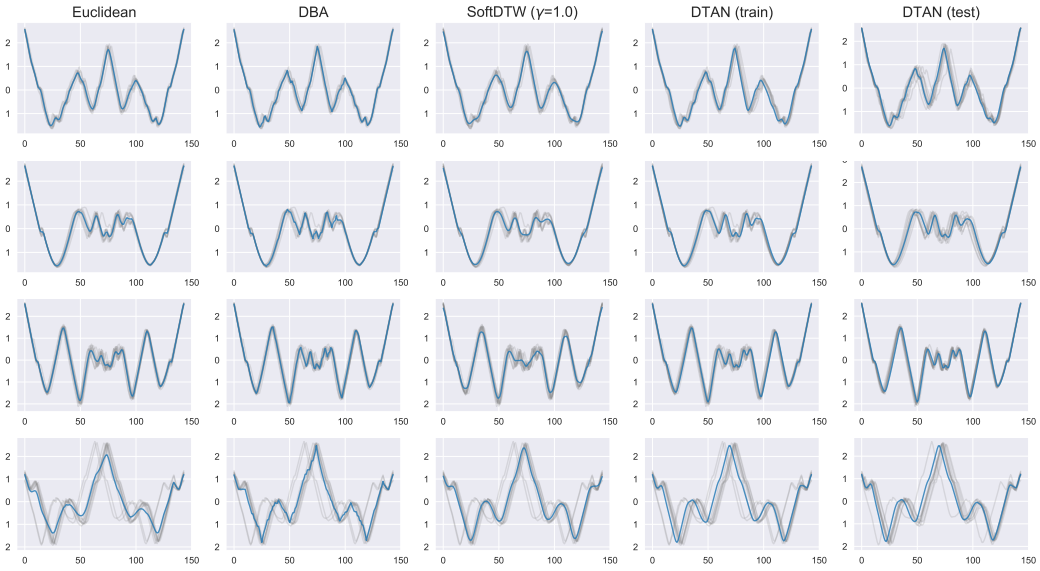


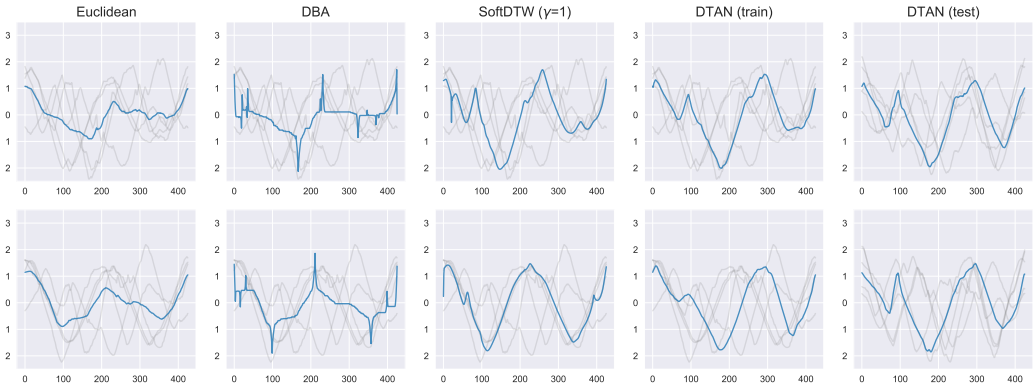
Figure 5: Within-class joint alignment of several datasets from the UCR archive [1]. The presented results are of previously-unseen test samples. The red and blue signals indicate the misaligned and aligned data average signal, respectively. The shaded area stands for $\pm$ standard deviation.

5.2 Average Signal / Barycenters Comparison

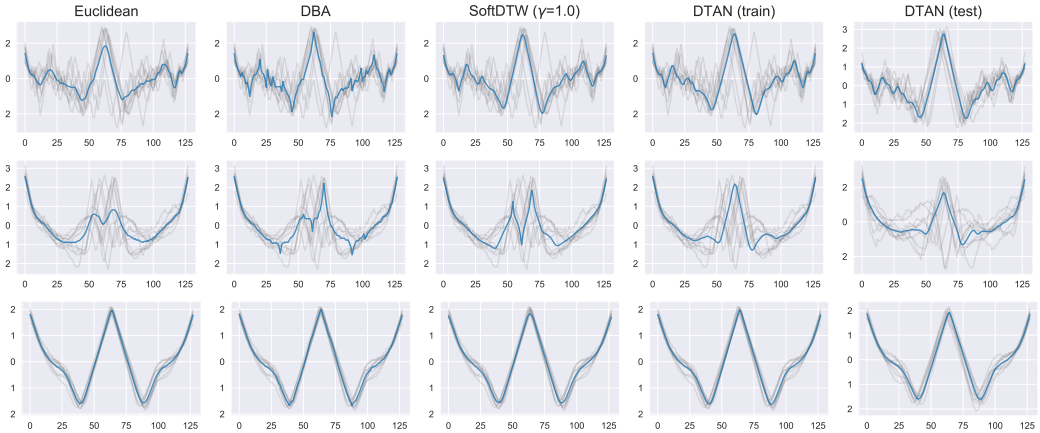
(a) Plane Dataset



(b) yoga Dataset



(c) SwedishLeaf Dataset



5.3 Recurrent DTAN

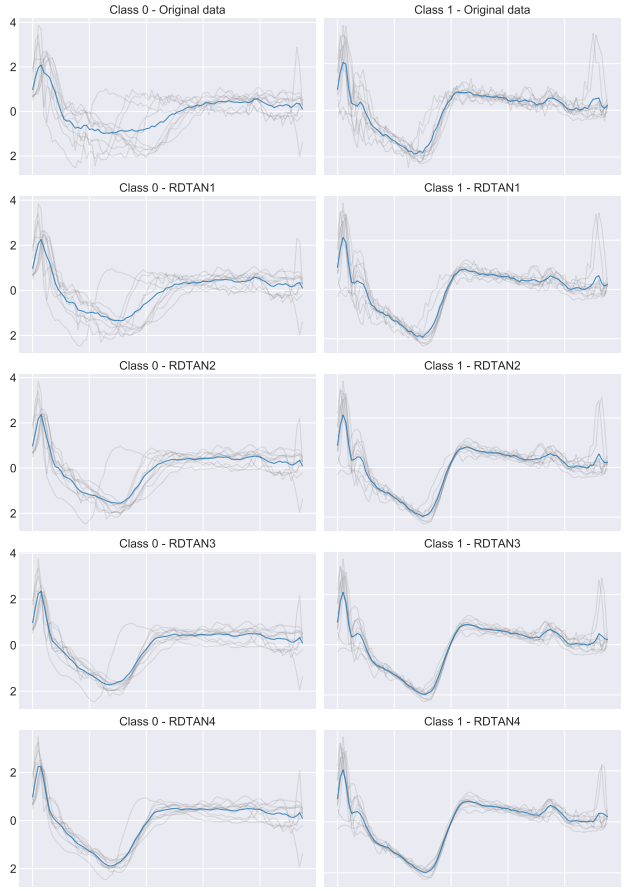


Figure 7: ECG200 dataset. Top row: 10 Randomly-chosen test samples from each class of the dataset. Second row until last: RDTAN joint alignment recurrent process output at each stage. The blue line indicates the sample mean of the signals.

6 Nearest Centroid Classification (NCC)

We show here detailed results of the NCC experiment mentioned in the paper ("StarlightCurves" dataset was excluded). For the baseline experiment we used the Euclidean mean of the misaligned set. We compare between DTAN and time-series barycenter-averaging methods: DTW Barycenter Averaging (DBA) and SoftDTW.

Table 2: Nearest Centriod Classificaiton results.

Dataset	**Baseline**	**DTAN**	**Softdtw**	**DBA**
50words	0.516484	**0.652747**	0.615385	0.615385
Adiac	0.549872	**0.695652**	0.501279	0.462916
ArrowHead	0.611429	**0.748571**	0.520000	0.474286
Beef	0.533333	**0.633333**	0.566667	0.400000
BeetleFly	0.850000	0.800000	0.850000	**0.900000**
BirdChicken	0.550000	**0.800000**	0.700000	0.600000
CBF	0.763333	0.914444	**0.971111**	0.965556
Car	0.616667	**0.816667**	0.683333	0.633333
ChlorineConcentration	0.333073	0.333073	**0.348177**	0.323698
CinC ECG torso	0.385507	**0.615942**	0.398551	0.445652
Coffee	0.964286	**1.000000**	0.964286	0.964286
Computers	0.416000	0.592000	**0.640000**	0.616000
Cricket X	0.238462	0.423077	**0.602564**	0.574359
Cricket Y	0.348718	0.541026	**0.571795**	0.541026
Cricket Z	0.305128	0.420513	**0.615385**	0.605128
DiatomSizeReduction	0.957516	**0.970588**	0.950980	0.950980
DistalPhalanxOutlineAgeGroup	0.817500	0.847500	**0.850000**	0.840000
DistalPhalanxOutlineCorrect	0.471667	0.471667	**0.490000**	0.488333
DistalPhalanxTW	0.747500	**0.780000**	0.760000	0.755000
ECG200	0.750000	**0.790000**	0.730000	0.720000
ECG5000	0.860444	**0.891333**	0.853778	0.834667
ECGFiveDays	0.689895	**0.977933**	0.670151	0.658537
Earthquakes	0.754658	0.773292	**0.822981**	0.574534
ElectricDevices	0.482687	0.534820	**0.539748**	0.538970
FISH	0.560000	**0.902857**	0.697143	0.651429
FaceAll	0.491716	0.804734	**0.827811**	0.796450
FaceFour	0.840909	0.829545	**0.852273**	**0.852273**
FacesUCR	0.539512	**0.857073**	0.812683	0.774634
FordA	0.495973	**0.604832**	0.552902	0.549570
FordB	0.499725	0.579758	**0.591309**	0.568482
Gun Point	0.753333	**0.880000**	0.733333	0.700000
Ham	0.761905	**0.790476**	0.733333	0.723810
HandOutlines	0.818000	**0.850000**	0.812000	0.804000
Haptics	0.392857	**0.457792**	0.373377	0.350649
Herring	0.546875	**0.703125**	0.609375	0.546875
InlineSkate	0.192727	**0.260000**	0.252727	0.232727
InsectWingbeatSound	**0.601010**	0.587374	0.328283	0.289394
ItalyPowerDemand	0.918367	**0.962099**	0.750243	0.730807
LargeKitchenAppliances	0.440000	0.482667	**0.733333**	0.728000
Lighting2	0.688525	**0.721311**	0.622951	0.639344
Lighting7	0.589041	0.712329	**0.726027**	0.698630
MALLAT	0.966738	**0.968870**	0.953945	0.952665
Meat	**0.933333**	**0.933333**	**0.933333**	0.916667
MedicalImages	0.385526	**0.468421**	0.461842	0.436842
MiddlePhalanxOutlineAgeGroup	0.732500	0.737500	**0.795000**	0.712500
MiddlePhalanxOutlineCorrect	**0.551667**	0.543333	0.495000	0.483333
MiddlePhalanxTW	0.591479	**0.596491**	0.581454	0.556391
MoteStrain	0.861022	**0.904153**	0.843450	0.826677
NonInvasiveFatalECGThorax1	0.769466	**0.853435**	0.710941	0.712977
NonInvasiveFatalECGThorax2	0.802036	**0.905344**	0.773028	0.763868
OSULeaf	0.359504	0.462810	**0.475207**	0.438017
OliveOil	**0.866667**	**0.866667**	0.800000	0.766667

Table 2: Nearest Centriod Classificaiton results.

Dataset	Baseline	DTAN	Softdtw	DBA
PhalangesOutlinesCorrect	0.625874	**0.642191**	0.637529	0.632867
Phoneme	0.078586	0.101793	**0.204641**	0.182489
Plane	0.961905	**1.000000**	0.990476	**1.000000**
ProximalPhalanxOutlineAgeGroup	0.819512	**0.853659**	**0.853659**	0.843902
ProximalPhalanxOutlineCorrect	0.646048	0.642612	**0.725086**	0.649485
ProximalPhalanxTW	0.707500	**0.817500**	0.747500	0.735000
RefrigerationDevices	0.354667	0.466667	**0.586667**	0.584000
ScreenType	0.442667	**0.445333**	0.389333	0.378667
ShapeletSim	0.500000	0.538889	**0.588889**	0.522222
ShapesAll	0.513333	**0.628333**	**0.628333**	0.603333
SmallKitchenAppliances	0.418667	0.621333	0.658667	**0.661333**
SonyAIBORobotSurface	0.811980	**0.893511**	**0.893511**	0.835275
SonyAIBORobotSurfaceII	0.793284	**0.811123**	0.772298	0.766002
Strawberry	0.668842	**0.843393**	0.649266	0.616639
SwedishLeaf	0.702400	**0.806400**	0.723200	0.681600
Symbols	0.864322	0.857286	**0.954774**	**0.954774**
ToeSegmentation1	0.574561	0.640351	**0.671053**	0.614035
ToeSegmentation2	0.546154	0.753846	**0.853846**	0.838462
Trace	0.580000	0.780000	**0.970000**	**0.970000**
TwoLeadECG	0.554873	**0.956102**	0.801580	0.811238
Two Patterns	0.464750	0.555750	**0.989750**	0.975000
UWaveGestureLibraryAll	0.849525	**0.920715**	0.833613	0.831937
Wine	0.555556	**0.574074**	**0.574074**	0.518519
WordsSynonyms	0.271160	**0.474922**	0.412226	0.344828
Worms	0.215470	0.259669	0.408840	**0.414365**
WormsTwoClass	0.541436	0.618785	**0.651934**	0.591160
synthetic control	0.916667	0.950000	**0.980000**	**0.980000**
uWaveGestureLibrary X	0.631212	0.681184	**0.706868**	0.676438
uWaveGestureLibrary Y	0.548297	**0.611669**	0.564768	0.525405
uWaveGestureLibrary Z	0.537409	**0.642099**	0.604132	0.592406
wafer	0.654445	**0.988968**	0.649416	0.511032
yoga	0.497000	**0.631667**	0.574000	0.557000

7 Alignment for CNN Classification

The alignment network (DTAN) was trained to minimize the within-class variance of each dataset (the training procedure is described in the main paper; "StarlightCurves" dataset was excluded). After the alignment phase is complete, DTAN output is fed into a subsequent classification network which is trained for time-series classification. During the classification training phase, the weights of the alignment net are frozen. The two classification networks (with and without the DTAN) are identical in terms of architecture and number of parameters.

Table 3: Comparison between DTAN-CNN and baseline CNN on the UCR archive. The results are the mean classification test accuracy and standard deviation (SD) of 5 runs per dataset.

Dataset	DTAN-CNN	SD	CNN	SD
50words	**0.7174**	0.013778	0.7060	0.007483
Adiac	**0.8266**	0.005953	0.7858	0.017174
ArrowHead	**0.6812**	0.037172	0.6788	0.053604
Beef	**0.2400**	0.064464	0.2066	0.067896
BeetleFly	**0.6200**	0.107703	0.5500	0.044721
BirdChicken	**0.6900**	0.124097	0.5400	0.048990
CBF	**0.8010**	0.090988	0.7154	0.108776

Table 3: Comparison between DTAN-CNN and baseline CNN on the UCR archive. The results are the mean classification test accuracy and standard deviation (SD) of 5 runs per dataset.

Dataset	**DTAN-CNN**	**SD**	**CNN**	**SD**
Car	**0.8800**	0.019380	0.7766	0.029159
ChlorineConcentration	0.7384	0.016439	**0.7890**	0.010334
CinCECGtorso	0.6644	0.126010	**0.6956**	0.044992
Coffee	**0.9286**	0.039072	0.8858	0.035273
Computers	**0.5384**	0.032259	0.5160	0.028955
CricketX	**0.6298**	0.011583	0.6268	0.021255
CricketY	**0.6732**	0.015197	0.6722	0.020054
CricketZ	**0.6230**	0.013609	0.6094	0.038119
DiatomSizeReduction	**0.9248**	0.020556	0.7006	0.196982
DistalPhalanxOutlineAgeGroup	**0.7454**	0.073826	0.7154	0.056220
DistalPhalanxOutlineCorrect	**0.8236**	0.010744	0.8188	0.004214
DistalPhalanxTW	0.7890	0.005586	**0.7904**	0.006499
ECG200	**0.8900**	0.017889	0.8780	0.019391
ECG5000	**0.8980**	0.025020	0.5330	0.230003
ECGFiveDays	**0.9012**	0.085775	0.7190	0.115331
Earthquakes	**0.8194**	0.001200	0.8130	0.008626
ElectricDevices	**0.6730**	0.005621	0.6722	0.009108
FISH	**0.9578**	0.005810	0.8698	0.026806
FaceAll	**0.7284**	0.005463	0.7156	0.005713
FaceFour	**0.7362**	0.107300	0.7136	0.055770
FacesUCR	**0.8888**	0.019477	0.8696	0.017636
FordA	0.8602	0.008931	**0.8742**	0.002482
FordB	**0.8168**	0.010962	0.8130	0.033923
GunPoint	**0.9548**	0.035561	0.8788	0.053462
Ham	**0.7162**	0.047008	0.6914	0.024589
HandOutlines	**0.8792**	0.008280	0.8584	0.012816
Haptics	**0.4986**	0.006437	0.4514	0.003137
Herring	**0.7096**	0.018467	0.5814	0.018282
InlineSkate	**0.3570**	0.011610	0.2106	0.050294
InsectWingbeatSound	0.6092	0.006400	**0.6344**	0.013749
ItalyPowerDemand	**0.9618**	0.003311	0.9368	0.015992
LargeKitchenAppliances	**0.5872**	0.004750	0.5446	0.016354
Lighting2	**0.7050**	0.023195	0.6330	0.039643
Lighting7	**0.6796**	0.025255	0.6190	0.042100
MALLAT	**0.8808**	0.030485	0.8416	0.028793
Meat	**0.8768**	0.022649	0.8034	0.235758
MedicalImages	**0.7278**	0.011479	0.7104	0.010651
MiddlePhalanxOutlineAgeGroup	0.7146	0.064769	**0.7440**	0.017297
MiddlePhalanxOutlineCorrect	**0.5544**	0.008163	0.5466	0.007060
MiddlePhalanxTW	**0.6154**	0.011056	0.6122	0.005036
MoteStrain	**0.8204**	0.011253	0.7600	0.098574
NonInvasiveFatalECGThorax1	**0.9188**	0.006242	0.8912	0.004534
NonInvasiveFatalECGThorax2	**0.9296**	0.003262	0.9224	0.002577
OSULeaf	**0.5924**	0.017614	0.5794	0.024377
OliveOil	**0.6800**	0.162823	0.5200	0.146969
PhalangesOutlinesCorrect	**0.7730**	0.011883	0.7414	0.007116
Phoneme	**0.0312**	0.008232	0.0272	0.010778
Plane	0.9904	0.008499	**0.9922**	0.011285
ProximalPhalanxOutlineAgeGroup	**0.7320**	0.140671	0.5454	0.103219
ProximalPhalanxOutlineCorrect	**0.9102**	0.010833	0.9032	0.010068
ProximalPhalanxTW	0.7898	0.034061	**0.8050**	0.010159
RefrigerationDevices	0.4208	0.033707	**0.4324**	0.027543
ScreenType	**0.4134**	0.021878	0.3830	0.019453
ShapeletSim	**0.5122**	0.023464	0.5056	0.013185
ShapesAll	**0.0248**	0.009621	0.0230	0.017989
SmallKitchenAppliances	**0.5468**	0.052055	0.4608	0.034219
SonyAIBORobotSurface	**0.6792**	0.033325	0.6190	0.128223
SonyAIBORobotSurfaceII	**0.7780**	0.030926	0.7382	0.058253
Strawberry	**0.9602**	0.008134	0.9478	0.018627

Table 3: Comparison between DTAN-CNN and baseline CNN on the UCR archive. The results are the mean classification test accuracy and standard deviation (SD) of 5 runs per dataset.

Dataset	**DTAN-CNN**	**SD**	**CNN**	**SD**
SwedishLeaf	**0.9488**	0.005600	0.9424	0.012500
Symbols	**0.6786**	0.047416	0.6372	0.075470
ToeSegmentation1	**0.6878**	0.037360	0.5808	0.066307
ToeSegmentation2	**0.7694**	0.009646	0.6430	0.061410
Trace	**0.9780**	0.011662	0.9040	0.041280
TwoLeadECG	**0.8558**	0.099494	0.6866	0.080822
TwoPatterns	**0.9964**	0.001497	0.9954	0.001497
UWaveGestureLibraryAll	0.9434	0.005238	**0.9440**	0.003847
Wine	**0.6444**	0.099468	0.5372	0.051219
WordsSynonyms	**0.6014**	0.003826	0.5762	0.022746
Worms	**0.4310**	0.033728	0.3836	0.130716
WormsTwoClass	**0.6188**	0.016485	0.5458	0.054551
syntheticcontrol	**0.8256**	0.004454	0.6254	0.192905
uWaveGestureLibraryX	**0.7748**	0.005706	0.7712	0.005418
uWaveGestureLibraryY	**0.6834**	0.003720	0.6832	0.003868
uWaveGestureLibraryZ	**0.7216**	0.007552	0.6812	0.005344
wafer	0.9922	0.001327	**0.9932**	0.001470
yoga	**0.8144**	0.009871	0.7722	0.008612

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